Mock Exam 1

Q1 Let the utility function of a consumer be

$$U(x, y) = x^{1/2}y^{1/4}.$$

a. Find the marginal utility of x and the marginal utility of y.
b. Derive the marginal rate of substitution of x for y, MRS_{xy}.
c. Determine whether the MRS is diminishing as x increases relative to y. Explain briefly.

Q2 Suppose x and y are related by the equation

$$x^2y + y^2 - 4x = 7.$$

a. Find $\frac{dy}{dx}$ using implicit differentiation.
b. Evaluate $\frac{dy}{dx}$ at the point $(1, 2)$.
c. State whether y is increasing or decreasing with respect to x at that point.

Q3 Let

$$f(x, y) = x^2y + 3xy^2.$$

a. Find the total differential df.
b. Use the total differential to approximate the change in f when x changes from 2 to 2.1, and y changes from 1 to 1.05.
c. Briefly explain the economic meaning of the total differential in a multivariable setting.

Q4 A consumer's utility function is

$$U(x, y) = 12x^{1/2} + 8y^{1/2} - x - 2y.$$

a. Find the first-order conditions.
b. Solve for the stationary point (x^*, y^*).
c. Use the second-order test to determine whether the stationary point is a maximum, minimum, or neither.

Q5 A firm's profit function is

$$\pi(K, L) = 16K^{1/4}L^{1/2} - 2K - L.$$

a. Find the first-order conditions for profit maximization.
b. Solve for the stationary point (K^*, L^*).
c. Use the second-order test to determine whether the stationary point is a maximum, minimum, or neither.

Mock Exam 2

Q1 Let the utility function be

$$U(x, y) = (x+1)^2(y+2)^3.$$

a. Find the marginal utility with respect to x and the marginal utility with respect to y.
b. Derive the marginal rate of substitution MRS_{xy}.
c. Comment on whether the willingness to substitute x for y depends on the consumption bundle.

Q2 Suppose x and y satisfy

$$x^2 + xy + y^2 = 12.$$

a. Find $\dfrac{dy}{dx}$ using implicit differentiation.
b. Find the slope of the curve at the point $(2, 2)$.
c. Determine the equation of the tangent line at that point.

Q3 Let

$$z = f(x, y) = x^3y^2 + 2xy.$$

a. Find the total differential dz.
b. Use the total differential to approximate the change in z when x changes from 1 to 1.02 and y changes from 2 to 1.97.
c. State clearly why the total differential is useful as an approximation tool.

Q4 A firm has the cost function

$$C(x, y) = \frac{1}{2}x^2 + xy + y^2 + 2x + y + 10.$$

a. Find the first-order conditions for stationary values of $C(x, y)$.
b. Solve for the stationary point (x^*, y^*).
c. Use the second-order test to determine whether the stationary point is a minimum, maximum, or neither.

Q5 A firm's profit function is

$$\pi(K, L) = 18K^{1/4}L^{1/2} - K - 2L.$$

a. Find the first-order conditions.
b. Solve for the stationary point (K^*, L^*).
c. Use the second-order test to determine whether the stationary point is a maximum, minimum, or neither.